

Linear Algebra

Question difficulty is indicated by 🐣 (lower difficulty), 🐣🐣 (greater difficulty), and 🐣🐣🐣 (greatest difficulty). A topic label is also given on the question.

Linear Transformation 🐣

Problem 1

SMMC 2021 B1 Let $n \geq 2$ be an integer, and let $\mathbf{O}$ be the $n \times n$ matrix whose entries are all equal to 0.

Two distinct entries of the matrix are chosen uniformly at random, and those two entries

are changed from 0 to 1. Call the resulting matrix $\mathbf{A}$. Determine the probability that $\mathbf{A}^2 = \mathbf{O}$, as a function of n .

Solution 1

Suppose the (i, j) - and the (p, q) -entries are turned from 0 to 1 to form the matrix $\mathbf{A}$. Note that $\mathbf{A}^2 = \mathbf{O}$ if and only if the dot product of any row of $\mathbf{A}$ with any column of $\mathbf{A}$ is equal to 0. This is equivalent to $\{i, p\} \cap \{j, q\} = \emptyset$.

Hence, if the two entries to be turned from 0 to 1 are picked one by one, the first one must be an entry not on the main diagonal. There are $n^2 n$ ways to choose this out of n^2 possibilities. If the first one is the (i, j) -entry with $i \neq j$, then the second entry must, in addition to not being on the main diagonal, not be in the i -th column or the j -th row.

Hence, out of the remaining $n^2 - 1$ possibilities, only $(n^2 - 1) - n - (2n - 3) = n^2 - 3n + 2$ will work; here, the ‘minus n ’ is for the diagonal entries, while the ‘minus $2n - 3$ ’ is for off-diagonal entries in row j union those in column i . It follows that the answer is

$$\frac{n^2 - n}{n} \times \frac{n^2 - 3n + 2}{n^2 - 1} = \frac{(n - 1)(n - 2)}{n(n + 1)}.$$

Linear Independence 🐣

Problem 2

Putnam 2003 Do there exist real polynomials $a(x), b(x), c(x), d(x)$ such that

$$1 + xy + (xy)^2 = a(x)c(y) + b(x)d(y)$$

holds identically?

Solution 2

Substitute $y = -1, 0, 1$, obtain 3-dimensional vector space, so impossible with 2 basis.

Linear Equations 🦉🦉

Problem 3

Let $A = (a_{ij})$ be a real $n \times n$ matrix satisfying

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

for all $1 \leq i \leq n$. Prove that A is invertible.

Solution 3

We know that A is invertible if and only if its columns are linearly independent. Suppose not, there would exist real numbers $c_1, \dots, c_n$, not all zero, such that

$$\sum_{j=1}^n c_j a_{ij} = 0$$

for each $i = 1, \dots, n$. In particular, let us pick i so that c_i has the largest absolute value among $c_1, \dots, c_n$. Then

$$a_{ii} = - \sum_{j \neq i} \frac{c_j}{c_i} a_{ij}.$$

Since $|c_i| \geq |c_j|$ for each j , we have

$$|a_{ii}| = \left| \sum_{j \neq i} \frac{c_j}{c_i} a_{ij} \right| \leq \sum_{j \neq i} \left| \frac{c_j}{c_i} \right| |a_{ij}| \leq \sum_{j \neq i} |a_{ij}| < |a_{ii}|$$

which is a contradiction. Therefore the columns of A are linearly independent, and hence A is invertible.

Linear Transformations and Eigenvector 🦉🦉

Problem 4

SMMC 2025 A2: Let p be a prime number. Given a square matrix M with entries in $\mathbb{Z}/p\mathbb{Z}$, an eigenvector of M is a non-zero vector $\vec{v}$ with entries in $\mathbb{Z}/p\mathbb{Z}$ such that $M\vec{v} = \lambda\vec{v}$ for some $\lambda \in \mathbb{Z}/p\mathbb{Z}$. For each positive integer n , determine the average number of eigenvectors of an $n \times n$ matrix with entries in $\mathbb{Z}/p\mathbb{Z}$. (Here $\mathbb{Z}/p\mathbb{Z}$ refers to the integers modulo p .)

Solution 4

This solution will cover the two easiest most intuitive methods:

Method 1:

1. Average = $\frac{|\{M: M\vec{v} = \lambda\vec{v}, \lambda \in \mathbb{Z}/p\mathbb{Z}\}|}{\#M}$.
2. $\#M = (p)^{n \times n}$ as p ways to fill in an entry of M .
3. Now for the numerator, fix an eigenvector $\vec{v}$ (Olympiad trick of "fixing") into the basis of M . The number of ways this can go is the number of ways to

chose the other $n - 1$ vectors to form the basis of M , chose the non-zero eigenvector and chose the eigenvalue.

4. There are $(p^n)^{n-1}$ to chose other basis vectors, p ways to chose a λ eigenvalue, and $p^n - 1$ ways to make a non-zero eigenvector (we remove $\vec{0}$ from the count).
5. So, $|\{M : M\vec{v} = \lambda\vec{v}, \lambda \in \mathbb{Z}/p\mathbb{Z}\}| = p^{n(n-1)}p(p^n - 1)$ which results in the Average $= \frac{p^n - 1}{p^{n-1}}$.

Method 2:

1. We rewrite the problem $M\vec{v} = \lambda\vec{v}$ into $(M - \lambda I)\vec{v} = \vec{0}$, which we can define as the solution of the linear map $T : M_{n \times n}(\mathbb{F}) \rightarrow \mathbb{F}^n : A\vec{v} = \vec{0}$ where $\mathbb{F}$ is $\mathbb{Z}/p\mathbb{Z}$.
2. In finding solutions to this linear map, we ignore the trivial $\vec{v} = \vec{0}$ as we are looking for non-trivial eigenvectors, so we assume full rank s.t. $\text{Rank}(A) = n$.
3. Consequently, by Rank-Nullity theorem $\text{Rank} + \text{Nullity} = \text{Dim}$, $\dim(\ker(T)) = \text{Nullity}(M) = \dim(M) - \text{Rank}(M) = n^2 - n$.
4. The number of ways to construct a matrix that satisfies this is $p^{n^2 - n}$ as that selects all the appropriate elements in the Kernel that meet the condition.
5. This times the number of ways to select an eigenvalue p , a non-zero vector $\vec{v}$ that becomes the eigenvector $p^n - 1$ gives $|\{A : A\vec{v} = \vec{0}, A = M - \lambda I, \lambda \in \mathbb{Z}/p\mathbb{Z}\}| = p^{n^2 - n}p(p^n - 1)$.
6. Since there are $p^{n \times n}$ of ways to select a matrix, the average number of eigenvector $= \{A : A\vec{v} = \vec{0}, A = M - \lambda I, \lambda \in \mathbb{Z}/p\mathbb{Z}\} / \#M = \frac{p^n - 1}{p^{n-1}}$.

Block Matrices and Determinant 🐾🐾🐾

Problem 5

Putnam 2018-A-2 Let $S_1, S_2, \dots, S_{2n-1}$ be the nonempty subsets of $\{1, 2, \dots, n\}$ in some order, and let M be the $(2^{n-1}) \times (2^{n-1})$ matrix whose (i, j) entry is

$$m_{ij} = \begin{cases} 0 & \text{if } S_i \cap S_j = \emptyset; \\ 1 & \text{otherwise.} \end{cases}$$

Calculate the determinant of M .

Solution 5

Let M_n be the matrix described in the problem. We will show that for all $n > 0$,

$$\det(M_{n+1}) = -\det(M_n)^2$$

Fix an ordering of $\text{Pow}(n) \setminus \emptyset$ as indicated in the problem. Then order the elements of $\text{Pow}(n+1) \setminus \emptyset$ as follows:

$$S_1, S_2, \dots, S_{2n-1}, N = \{n+1\}, S_1 \cup N, S_2 \cup N, \dots, S_{2n-1} \cup N$$

Then the matrix M_{n+1} may be described in blocks as

$$\begin{pmatrix} M & O_c & M \\ O_r & 1 & J_r \\ M & J_c & JJ \end{pmatrix}$$

where O_r is a row of zeros, J_r is a row of 1s, O_c and J_c are columns of 0's and 1's, and JJ is a square matrix filled with 1s.

Subtract the first $2^n - 1$ rows from the last $2^n - 1$ rows in the obvious pairs to zero out the lower-left corner. The matrix JJ will be replaced by the complement of M .

Then subtract the middle row from all the rows below it; J_c will be replaced by O_c , and the lower right corner will be replaced by $-M$.

Now the matrix is in block-diagonal form, so its determinant is $\det(M) \cdot 1 \cdot \det(-M)$. Since $2^n - 1$ is odd, $\det(-M) = -\det(M)$ and we are done.

In the case $n = 1$, M is just the 1×1 matrix $M = (1)$ whose determinant is 1. By the preceding we then see by induction that $\det(M) = -1$ for all $n > 1$.

Powers of Matrices and Nil-potency 🐾🐾🐾

Problem 6

Putnam 2012-A-5 Let $\mathbb{F}_p$ denote the field of integers modulo a prime p , and let n be a positive integer.

Let v be a field vector in $\mathbb{F}_p^n$ and let M be an $n \times n$ matrix with entries in $\mathbb{F}_p$, and define $G : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ by $G(x) = v + Mx$. Let G^k denote the k -fold composition of G with itself, that is $G^1(x) = G(x)$ and $G^{(k+1)}(x) = G(G^k(x))$.

Determine all pairs p, n for which there exist v and M such that the p^n vectors $G^k(0), k = 1, 2, \dots, p^n$ are distinct.

Solution 6

The pairs (p, n) with the specified property are those pairs with $n = 1$, together with the single pair $(2, 2)$. We first check that these do work. For $n = 1$, it is clear that taking $v = (1)$ and $M = (0)$ has the desired effect.

For $(p, n) = (2, 2)$, we take $v = \begin{pmatrix} 0 & 1 \end{pmatrix}$ and $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and then observe that

$$G^k(0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad k = 0, 1, 2, 3.$$

We next check that no other pairs work, keeping in mind that the desired condition means that G acts on $\mathbb{F}_p^n$ as a cyclic permutation. Assume by way of contradiction that (p, n) has the desired property but does not appear in our list. In particular, we have $n \geq 2$.

Let I be the $n \times n$ identity matrix over $\mathbb{F}_p$. Decompose $\mathbb{F}_p^n$ as a direct sum of two subspaces V, W such that $M - I$ is nilpotent on V and invertible on W . Suppose that $W \neq 0$. Split v as $v_1 + v_2$ with $v_1 \in V, v_2 \in W$. Since $M - I$ is invertible on W ,

there exists a unique $w \in W$ such that $(M - I)w = -v_2$. Then $G^{(k)}(w) - w \in V$ for all nonnegative integers k . Let k be the least positive integer such that $G^{(k)}(w) = w$; then k is at most the cardinality of V , which is strictly less than p^n because $W \neq 0$. This gives a contradiction and thus forces $W = 0$.

In other words, the matrix $N = M - I$ is nilpotent; consequently, $N^n = 0$. For any positive integer k , we have

$$\begin{aligned} G^{(k)}(0) &= v + Mv + \cdots + M^{k-1}v \\ &= \sum_{j=0}^{k-1} \sum_{i=0}^{n-1} \binom{j}{i} N^i v \\ &= \sum_{i=0}^{n-1} \binom{k}{i+1} N^i v. \end{aligned}$$

If $n \geq 2$ and $(p, n) \neq (2, 2)$, then $p^{n-1} > n$ and so $G^k(0) = 0$ for $k = p^{n-1}$ (because all of the binomial coefficients are divisible by p). This contradiction completes the proof.